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Bacillus subtilis impact on plant growth, soil health and environment: Dr. Jekyll and Mr. Hyde

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Abstract

The increased dependence of farmers on chemical fertilizers poses a risk to soil fertility and ecosystem stability. Plant growth-promoting rhizobacteria (PGPR) are at the forefront of sustainable agriculture, providing multiple benefits for the enhancement of crop production and soil health. *Bacillus subtilis* is a common PGPR in soil that plays a key role in conferring biotic and abiotic stress tolerance to plants by induced systemic resistance (ISR), biofilm formation and lipopeptide production. As a part of bioremediating technologies, *Bacillus* spp. can purify metal contaminated soil. It acts as a potent denitrifying agent in agroecosystems while improving the carbon sequestration process when applied in a regulated concentration. Although it harbours several antibiotic resistance genes (ARGs), it can reduce the horizontal transfer of ARGs during manure composting by modifying the genetic makeup of existing microbiota. In some instances, it affects the beneficial microbes of the rhizosphere. External inoculation of *B. subtilis* has both positive and negative impacts on the endophytic and semi-synthetic microbial community. Soil texture, type, pH and bacterial concentration play a crucial role in the regulation of all these processes. Soil amendments and microbial consortia of *Bacillus* produced by microbial engineering could be used to lessen the negative effect on soil microbial diversity. The complex plant–microbe interactions could be decoded using transcriptomics, proteomics, metabolomics and epigenomics strategies which would be beneficial for both crop productivity and the well-being of soil microbiota. *Bacillus subtilis* has more positive attributes similar to the character of Dr. Jekyll and some negative attributes on plant growth, soil health and the environment akin to the character of Mr. Hyde.

KEYWORDS

antibiotic resistance genes, *Bacillus* consortia, PGPR, plant-microbe interaction, sustainable agriculture

INTRODUCTION

The excessive use of chemical fertilizers in current agricultural practices deteriorates soil quality, contaminates groundwater and poses a risk to biodiversity (Ramakrishna et al., 2019). Continuous and indiscriminate use of chemicals adversely affects human and animal health by accumulating in the crops and entering the food chain through the biomagnification process (Naik et al., 2019). Considering all the scenarios of modern

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agricultural practices, organic farming, and traditional agricultural methods, the use of biofertilizers is essential for a sustainable agroecosystem.

Healthy soil maintenance depends upon certain parameters like carbon content, water filtration ability, oxygen holding capacity, erosion minimization and microbial species diversity (Parray & Shameem, 2020). Chemical fertilizers induce plant growth but negatively impact all the above factors resulting in loss of soil fertility. Application of plant growth-promoting rhizobacteria (PGPR) in agriculture is an alternative to reduce or replace chemical fertilizers and improve soil fertility (Ramakrishna et al., 2019). PGPR can be exploited as an efficient tool to induce plant growth in stress conditions. They can interact with plants directly for enhancement of nutrient uptake (macro and micronutrients), phytohormone production, regulation of stress hormones and enzymes such as ethylene, abscisic acid (ABA), 1-aminocyclopropane 1-carboxylase (ACC) deaminase, and indirectly by biocontrol activity against pathogens through intraspecific interaction, cell wall lysing enzyme secretion, lipopeptides production and induced systemic resistance (ISR) (Ramakrishna et al., 2020; Yadav et al., 2020; Yadav et al., 2021).

Bacillus subtilis is a common PGPR present in rhizosphere. The spores produced by *Bacillus* are long-lived and sustain several harsh environmental conditions; thus can be employed in diverse growth conditions (Nicholson et al., 2000). It is one of the prominent biofilm-forming rhizobacteria, which help plants in nutrient uptake and root colonization (Figure 1). The use of *Bacillus* inoculants in farmlands reduces the release of nitrogen and ammonia gases (Sun et al., 2020a). It plays a crucial role in the regulation of the biogeochemical cycles providing soil with a proper aeration capacity to maintain its ecosystem. The presence of membrane efflux pumps and BceAB type transporters in *B. subtilis* provide intrinsic resistance against a broad range of antibiotics (Rismondo & Schulz, 2021). Though it can reduce the concentration of other antibiotic resistance genes (ARGs) and mobile genetic elements from composting manure, its higher concentration affects other beneficial properties. Very little information is present about the horizontal transfer of ARGs and their significance in agriculture. Although *B. subtilis* is considered a beneficial microbe, sometimes it affects the endophytic and rhizospheric microbiota negatively by decreasing their number and modifying the genetic makeup (US EPA, 1997). It depends on soil texture, composition, pH and bacterial concentration in the rhizosphere. Therefore, it is essential to perform a feasibility study before applying microbial inoculants in the soil. Several omics technologies (genomics, epigenomics, transcriptomics, proteomics, metabolomics) are available for the functional and structural assessment and characterization of bacterial diversity. Preparation of microbial inoculants by genetic engineering with no effect on other microbial communities would benefit both the plant and microbe. Even after the successful application of microbial consortia as biofertilizers during experimental trials, some of them fail to reach farmer's satisfaction in the field (Soumare et al., 2020). Thus, proper storage and transport of biofertilizers are required for increasing their field efficiency and shelf life. The objective of this review is to

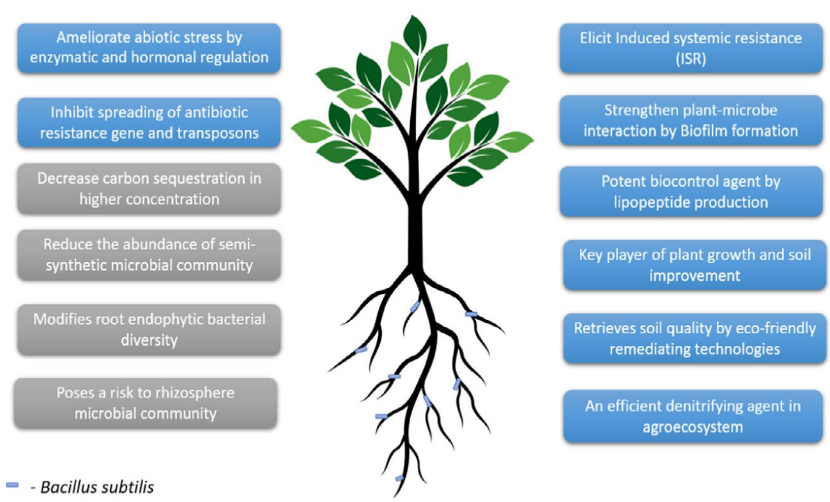

FIGURE 1 Impact of *Bacillus subtilis* on plant growth, soil health improvement and its environmental effects. *Bacillus subtilis* regulates the stress responses of plants by several hormonal and enzymatic mechanisms where induced systemic resistance (ISR), lipopeptide and biofilm formation. It plays a vital role in soil remediation and as a denitrifying agent in agriculture. It reduces the spread of antibiotic resistance genes and mobile genetic elements. Even after having a plethora of positive effects, *B. subtilis* has some negative impact on the rhizosphere and endophytic microbial community, which cannot be ignored. The concentration of bacteria plays an important role in maintaining the biogeochemical cycles like carbon cycle and its sequestration process

highlight both the positive and negative effects of *B. subtilis* taking into account recent research in this field and recommendations for a future road map for their efficient application in enhancing crop productivity.

KEY PLAYER IN PLANT GROWTH PROMOTION AND NUTRIENTS ACQUISITION

Growth-promoting traits

Plants encounter both biotic and abiotic stress in different stages of their lifetime. Studies have revealed that inoculation of certain bacteria instead of chemical fertilizers would be beneficial for both the plant and microbe world (Ramakrishna et al., 2020). *Bacillus subtilis*, a plant growth-promoting bacterium, acts synergistically with the host plant in regulating its physiochemical processes by producing different volatile organic compounds (VOCs). Using gas chromatography–mass spectroscopy analysis, scientists have observed that VOCs like albuterol and 1,3-propanediol, produced by *B. subtilis* SYST2, mediate plant growth promotion by upregulating the photosynthetic activity and phytohormone production (Tahir et al., 2017). The endospore-producing *Bacillus* spp. tend to survive for a longer period in soil and promote plant growth more efficiently compared to other plant growth-promoting bacteria (PGPB) that do not produce endospores (Yadav et al., 2020, 2021). Furthermore, it was reported that *Bacillus* spp. improves the photosynthetic capacity of wheat plants as they produce siderophores which chelate iron and supply iron required for photosynthetic machinery. *Bacillus* spp. reduce the iron deficiency in plants and mitigate the heavy metal induced oxidative stress as they protect the indole-3-acetic acid (IAA) from oxidative damage (Ferreira et al., 2019). Previously, it was documented that siderophore producing *Bacillus* spp. in combination with mycorrhizal fungi enhance the nutrient acquisition in wheat grain and root tissues (Yadav et al., 2021). Siderophore produced by these microbes in the rhizosphere solubilize the unavailable forms of various nutrients via acidification. Furthermore, plants acquire the nutrients from the nutrient–siderophore complexes using root-mediated processes like chelate degradation or ligand exchange reaction (Rajkumar et al., 2010).

Abiotic stress has a negative effect on plant growth and development. *Bacillus subtilis* has a high resilience towards abiotic stress due to its genomic organization and metabolic capabilities (Table 1). It harbours numerous genes for plant growth promotion and abiotic stress control (Leontidou et al., 2020). Spx acts as the central regulator of stress control by binding to the alpha subunit of RNA polymerase (Schäfer & Turgay, 2019). In addition, alarmones, secondary messengers of nutrient starvation, act as efficient intermediates to counter heat shock and other stresses (Schäfer et al., 2020). Endophytic species are more effective as compared to epiphytic ones. Inoculation of arbuscular mycorrhizal fungi (AMF) along with *B. subtilis* in the Talh tree has shown a synergistic effect in ameliorating the adverse effects of salinity stress on plant metabolism by producing glycine, betaine and proline (Hashem et al., 2016). *Bacillus* spp. harbour several genes' *KatA, SodA, trxA* and *perR* associated with protecting the plants from oxidative stress (Zubair et al., 2019). Similarly, *Bacillus* spp. induce the expression of *DegS, dpsU20, desk, desR* and ResD which mitigates cold stress in plants through signal transduction pathways. Several osmotic regulatory genes *OpuAC* and *ohR* were identified in psychotropic *Bacillus* strains which have a role in maintaining osmotic homeostasis in plants. They also improve plant survival under abiotic stress by inducing the expression of peroxidase (POD) enzymes thereby minimizing the negative effect caused by adverse agroclimatic conditions. Thus, regulating abiotic stress responses of plants using *Bacillus* species could improve global agricultural productivity.

Phytohormones

Bacillus spp. act as biostimulants through the production of phytohormones, auxin and cytokinin as well as expansin that contribute to plant growth and development (Zubair et al., 2019). The presence of ACC deaminase genes in *Bacillus* spp. confers the ability to lower the ethylene level in plants thereby enhancing growth and drought tolerance in treated plants (Gowtham et al., 2020). Similarly, VOCs such as 2-ethyl hexanol, tetrahydrofuran-3-ol and 2-heptanone belonging to alcohols, ketones, furans, terpenes and sulphur compounds elicit plant growth through interaction with multiple compounds (Jiang et al., 2019). VOCs emitted by *Bacilllus* spp. significantly enhanced the endogenous auxin level and decreased the strigolactones level indicating that VOCs act as signalling molecules modulating the regulatory pathways in associated plants. The crosstalk between auxin and strigolactones (SLs) promotes plant growth induced by VOCs emitted by *Bacillus* spp. Cytokinins (CK) produced by *Bacillus* spp. are associated with cell division, photomorphogenic development and shoot differentiation. Likewise, gibberellins play a significant role in seed germination, flower initiations and fruit development. *Bacillus* spp. could be exploited as an alternative to obtain these phytohormones to enhance agricultural crop productivity. These microbes produce the CKs, zeatin (Z), zeatin nucleotide (ZN) such as izopentyl

TABLE 1 *Bacillus subtilis* mediated abiotic stress tolerance in plants

Abiotic stress	Strain	Plant	Mechanism	Reference
Drought	B26	*Brachypodium distachyon*	Upregulation of drought response genes DREB2B like; modulation of DNA methylation genes	Gagné-Bourque et al. (2015)
Drought	B26	Timothy	Accumulation of osmolyte in shoot and root (GABA)	Gagné-Bourque et al. (2016)
Drought	RJ14	*Pisum sativum* *Vigna mungo*	Upregulation of ACC deaminase; downregulation of ACC oxidase gene	Saikia et al. (2018)
Drought	....	Chick pea (*Cicer arietinum)*	Significant accumulation of nicotinamide and 4-hydroxy methyl glycine	Khan et al. (2019)
Drought	HAS31	Potato	Maintenance of catalase, peroxidase and superoxide dismutase enzymes	Batool et al. (2020)
Drought	10-4	Wheat	Decrease in lipid peroxidation; electrolyte leakage from tissues	Lastochkina et al. (2020)
Salt	EGY16	*Thymus vulgaris*	Decrease concentration of antioxidant enzymes	Abdelshafy Mohamad et al. (2020)
Salt	GB03	White clover	Regulation of plant chlorophyll content, cell membrane integrity, leaf osmotic potential, ionic balance	Han et al. (2014)
Salt	10-4	Wheat	Slow down statolite starch hydrolysis	Lastochkina et al. (2017)
Salt	SU47	Wheat	Reducing the concentration of antioxidant enzymes	Upadhyay et al. (2012)
Salt	BERA 71	*Acacia gerradii* Benth.	Modulating osmotolerant system and antioxidant enzyme system	Hashem et al. (2016)
Salt	NRCB003	*Medicago sativa L.*	Increasing ACC deaminase activity	Zhu et al. (2020)
Salt	QST-713	Tomato	-------------	Medeiros and Bettiol (2021)
Lead (Pb) Toxicity	FBL-10	*Solanum melongena*	Activate antioxidant system of plants; decrease the level of peroxide and malondialdehyde (MDA)	Shah et al. (2021)
Iron (Fe) deficiency	STU-6	Tomato	Enhancement of polyamine mediated iron remobilization	Zhou et al. (2019)
Cadmium stress	MF497446	Cow pea	Produce IAA, GA and siderophores to induce plant growth under Cd stress	El-Nahrawy et al. (2019)
Cd, Cr, Ni	KP717559	*Brassica juncea*	Enhance phytoextraction by plants while producing IAA, GA, HCN for plant growth	Ndeddy Aka and Babalola (2016)
As, Cd, Cu, Ni	KUJM2	*Lens culinaris*	Modulate translocation /retention resulting in lowered toxicant levels in plant parts	Mondal et al. (2019)
Ni, Pb, Cd	---------	Chick pea	Increase proline content and decrease lipid peroxidation	Khan and Bano (2018)
Pb toxicity	PBRB3	Mung bean	Higher antioxidative enzyme activity	Arif et al. (2019)
Pb toxicity	CIK512	Radish	Regulating the homeostasis of anti-oxidants	Ahmad et al. (2018)
Osmotic stress tolerance	-----------	Arabidopsis	Higher proline production	Chen et al. (2007)
Heat stress	-----------	Wheat	Enhanced activity of antioxidant, proline	Ashraf et al. (2019)

adenosine (iPA), izopentyl adenine (iP), and zeatin riboside (ZR) that promotes plant growth and shoot initiation in associated plants (Poveda & González-Andrés, 2021). Z and ZR producing *Bacillus* spp. enhance the root exudation of amino acids in wheat which positively reshape the neighbouring soil microbial community (Kudoyarova et al., 2014). Similarly, *Bacillus* spp. produce gibberellic acid (GA), which is associated with enhanced seed germination and stem elongation as reported in common alder (*Alnus glutinosa*). GA-producing *Bacillus* spp. positively correlated with increased macro- and micronutrients, fructose, carotenoids and gamma-aminobutyric acid (Govindasamy et al., 2010; Nascimento et al., 2020). Furthermore, GAs produced by *Bacillus* spp. improved the thermotolerance in wheat and soybean by inducing the expression of salicylic acid (SA) and jasmonic acid (JA) and downregulating ABA (Kang et al., 2019; Oleńska et al., 2020). The multitude of phytohormones produced by *B. subtilis* and *Bacillus* spp. explain their positive effects on the enhancement of plant growth and nutrient content (Poveda & González-Andrés, 2021; Saxena et al., 2020).

Biocontrol and stimulation of immunity

The traditional method of disease triangle by plant pathogens includes host susceptibility, pathogen virulence and environmental conditions (Scholthof, 2007). *Bacillus* spp. affect this disease triangle by direct and indirect pathways (Radhakrishnan et al. (2017). The indirect method includes the formation of biofilm, plant growth promotion, competition for space and nutrients, and ISR. In the direct method of disease control, *B. subtilis* produce various lipopeptides, cell lytic enzymes, antioxidants and hormones which affect a wide range of fungi and bacteria. Surfactin, iturin and fengycin are the major lipopeptides produced, which have a tremendous role in regulating a plethora of phytopathogens (Table 2). Surfactin is one of the prominent lipopeptides produced by *B. subtilis*. It is a cyclic lipoheptapeptide, consisting of three beta-hydroxy fatty acyl chains and seven amino acid residues (Wang et al., 2014). Surfactin shows antifungal activity against a wide range of fungal phytopathogens by inducing pore and ion channels through the lipid bilayer, cell cycle arrest and apoptosis. Fengycin is a cyclic lipodecapeptide having 16–19 carbons of beta-hydroxy fatty acid, which upon changing its length and branching pattern produces different isoforms (Ntushelo et al., 2019). It targets the phospholipase A2 enzyme and disrupts the fungi cell wall. A higher concentration of fengycin ruptures the cell membrane and forms permanent lesions affecting cellular integrity. It elicits systemic resistance in plants and is effective against several genera of fungi. Iturins are amphiphilic compounds having a heptapeptide backbone and beta-hydroxy fatty acid chain of 17–19 carbon atoms. Iturins target the fungi cells by ion pore mediating cell wall disruption. They change the morphological structure of conidia and affect hyphae formation in *Fusarium* spp. *Bacillus subtilis* produces many hydrolytic enzymes, proteases, cellulase and glucanase, which reshape the rhizosphere environment so that it is more favourable for plant growth (Miljaković et al., 2020). Different *Bacillus* strains produce stable biofilms which act as a good biocontrol agent against fungal pathogens. Similarly, ribosomal peptide antibiotic such as bacteriocins secreted by *Bacillus* spp. induce the innate host immunity (Lee & Kim, 2011). Lipopeptide-producing *Bacillus* strains form biofilms around plant roots thereby augmenting the antimicrobial activity in rhizosphere soil. Furthermore, the lipopeptides produced by *Bacillus* have a plethora of beneficial applications in both industry and agroecosystem. In some cases, the lipopeptides can induce a negative effect on soil microbes. In addition, the VOCs and phytohormones produced by *B. subtilis* might affect endophytic and epiphytic microbiota. These beneficial microbes which are a boon for the plant rhizospheric ecosystem may act as a curse for animal gut microbiota when ingested. *Bacillus subtilis* forms its own ecosystem through biofilm formation and produces several toxins to fight against pathogenic organisms. However, they may also affect the good microbes as they are trapped in its ecosystem. There are a lot of hidden secrets in the plant rhizosphere that need to be explored in a timely manner. A comprehensive understanding is required to avoid damage to plant and animal diversity.

In natural conditions, plants are exposed to a large range of organisms like insects and micro-organisms (biotrophs, hemi-biotrophs and necrotrophs). To cope with the adverse effects of pathogen infection, plants activate a layer of defence apparatus depending upon the nature and lifestyle of the pathogen (Nguyen et al., 2020). The 'Molecular associated membrane repeats' (MAMPS) along with the pattern-recognition receptors (PRRs) play a crucial role in recognizing the microbial molecules, thereby developing a particular defence response. The timing and type of immune response characterize its nature, as the secondary ones are stronger as compared to the primary response. Systemic acquired response (SAR) and ISR are likely to be the second defence responses in plants, mediated by different phytopathogens (Figure 2). Once a pathogen infects the plant, it reduces its energy metabolism and triggers the defence mechanism for developing a long-lasting SAR. It has been shown that treatment of plant roots with certain PGPB evolve such a protective effect on plants which can also protect the above-ground plant parts from a diverse scale of harmful microbes. This physiological state increases the defence capacity of plants simplified as ISR and the process is known as priming (Choudhary et al., 2007). After getting the pathogen-related signal, these primed plants upregulate the generation of reactive oxygen species (ROS), defence-related gene expression, callose deposition in the cell walls and collection of antimicrobial phytoalexins (Nguyen et al., 2020). *Pseudomonas* spp. increase the basal defence mechanism of plants against multiple soil pathogens by ISR (Romera et al., 2019). As per recent reports, *Bacillus* spp. are in the front line of ISR induction while stimulating plant growth in a collective manner. It has been reported to protect plants against

TABLE 2 Role of *B. subtilis* in plant disease control by lipopeptide production

S. No.	Strains	Host	Pathogen	Mechanism	Reference
1.	EA-CB0015	Chrysanthemum	*Botrytis cinera* *C. acutatum*	Iturin A Fengycin c	Arroyave-Toro et al. (2017)
2.	9407	Watermelon	*Acidovorax citrulli* (Bacterial fruit blotch)	Surfactin reduces biofilm formation & swarming motility	Fan et al. (2017)
3.	V26	Potato	*Fusarium oxysporum* *Fusarium solani*	Surfactin Fengycin Iturin Bacillomycin	Ben Khedher et al. (2021)
4.	SL-6	Pome	*Alternaria alternata* *Botrytis cinera*	Surfactin Iturin Fengycin	Cozzolino et al. (2020)
5.	ATCC21556	Lettuce	*Fusarium oxysporum*	Iturin A	Fujita and Yokota (2019)
6.	QST 713	Cucumber	Powdery mildew Fusarium rot	Fengycin	Punja et al. (2019)
7.	ET-1	Lemon Strawberry	Green and gray mould (*P. digitatum*; *B. cinerea*)	Iturin A	Ambrico & Trupo, 2017
8.	9407	Apple	Apple rot (*Botryosphaeria dothidea*)	Fengycin inhibits mycelial growth	Fan et al. (2017)
9.	NCD-2	Chinese cabbage	Clubroot (*Plasmidophora brassicae*)	Fengycin	Guo et al. (2019)
10.	Q-11	Apple	Blue mould (*Penicillium expansum*)	Iturin Fengycin inhibits conidia germination	Rodríguez-Chávez et al. (2019)
11.	MBI 600	Cucumber	Gray mould (*Botrytis cinera)*	Fengycin	Samaras et al. (2021)
12.	WL-2	Potato	Potato late blight (*Phytophthora infestans*)	Iturin (Oxidative stress, cell wall disrupture)	Wang et al. (2020)
13.	NCD-2	Wheat	*Rhizoctonia solani AG-8*	Surfactin Iturin Fengycin	Zhang et al. (2021)
14.	FZV-1	Pea	*Fusarium solani*	Surfactin Fengycin Siderophores	Riaz et al. (2021)
15.	GLB191	Grape	Grapevine downy mildew (*oomycete Plasmopara viticola)*	Surfactin Fengycin	Li et al. (2019)

bacterial stem blight, crown rotting and leaf spotting fungal pathogens, blue mould, nematodes and several wilt-causing pathogens (Jinal & Amaresan, 2020). SAR is phenotypically similar to ISR but the signalling pathways underlying it are different. SAR relies on the salicylic acid signalling pathway and accumulation of pathogenesis-related proteins (PR genes) unlike ISR, which is elicited by the ethylene/jasmonate signalling pathway and the regulatory gene NPR1 (Dimopoulou et al., 2019). Abscisic acid (ABA) has been shown to play a key role in ISR induction as ABA-deficient strains allowed stomatal closure after treatment with *B. subtilis* FB17 (Kumar et al., 2012). It activates ABA and salicylic acid signalling pathways to restrict pathogen entry through stomata.

Bacterial lipopeptides can elicit ISR in bean mediated by surfactin and fengycin produced by *B. subtilis* S499 (Ongena et al., 2007). The same result was shown by *B. subtilis* CtpxS2-1 in Andean lupin (Yánez-Mendizábal & Falconí, 2021). Other pieces of evidence show that bacterial VOCs such as 2,3-butanediol, induce ISR in plants through salicylic acid/ethylene signalling pathway and PR gene upregulation (Farag et al., 2013). In some plants like tomato, *B. subtilis* CBR05 elicits the ISR response along with the biosynthesis of vitamin B in the de novo pathway

(Chandrasekaran et al., 2019). Certain natural elicitors such as 3,4-dihydroxy-3-methyl-2-pentanone are also responsible for ISR as observed in the case of the PGPR *B. subtilis* HN09, which dominates HN09- ISR (Liu et al., 2019).

Biofilm production contributing to rhizocompetence

Biofilms produced by microbes constitutes a mixture of species, it makes molecular genetic studies of a single species quite difficult (Arnaouteli et al., 2021). To overcome this difficulty, in vitro biofilm making is being carried out. *Bacillus subtilis* biofilms are predominantly being used with the ancestral strain NCIB3610 and the three commonly used experimental systems are pellicle formation, where the complex bacteria form an interface between air and liquid, rugose colony formation where the biofilm is formed on semisolid agar surfaces (Branda et al., 2001) and the formation of biofilm on plant roots using in vitro agricultural settings (Cairns et al., 2014). Time-resolved analysis of *B. subtilis* pellicle biofilm formation identified metabolic changes during the process like increasing the level of fatty acid degradation than biosynthesis, upregulation of iron metabolism, conversion from acetate to acetoin fermentation and enhanced Krebs cycle activity during the early phase (Pisithkul et al., 2019). The metabolomic, transcriptomic and proteomic analyses has indicated that the complex metabolic remodelling of biofilm formation is highly regulated at the transcriptional level. *Bacillus subtilis* uses quorum sensing (QS) for biofilm production by secreting autoinducer signalling molecules into the extracellular medium (Kalamara et al., 2018). ComQXPA proteins regulate the QS system by phosphorylation and dephosphorylation. For instance, phosphorylation of ComA in the cytoplasm induces surfactin production.

Biofilm gives the microbes a sessile lifestyle from free moving planktonic life where they can take several advantages like getting nutrition easily from root exudates and surviving from environmental fluctuations. Shreds of evidence suggest that citric acid of cucumber root exudates induce biofilm formation in *Bacillus amyloliquefaciens* SQR9 and fumaric acid of banana root exudates had a significant role in chemotaxis and biofilm formation of *B. subtilis* N11 (Zhang et al., 2014). These microbial biofilms help the plants in different ways. Inoculation of biofilm-forming PGPR like *B. subtilis* enhances crop yield through different plant growth mechanisms. Plant growth regulators and antimicrobials produced by such PGPR regulate the plant metabolic pathways in a positive way. Toxins produced by *B. subtilis* such as YIT toxin through yitPOM operon act as a good competitor for other microbes serving as a biocontrol agent for the plant (Kobayashi & Ikemoto, 2019). These can be used as good biofertilizers by inoculating with nitrogen-fixing microbes (Seneviratne et al., 2010). This biofilm formation in the root rhizosphere also helps the plant in the management of its biotic and abiotic stress balance. The biofilm-forming activity of *B. amyloliquefaciens* has been shown to enhance the salt stress tolerance of barley (Kasim et al., 2016). Several bacterial metabolic pathways require biofilm formation for their optimal activity.

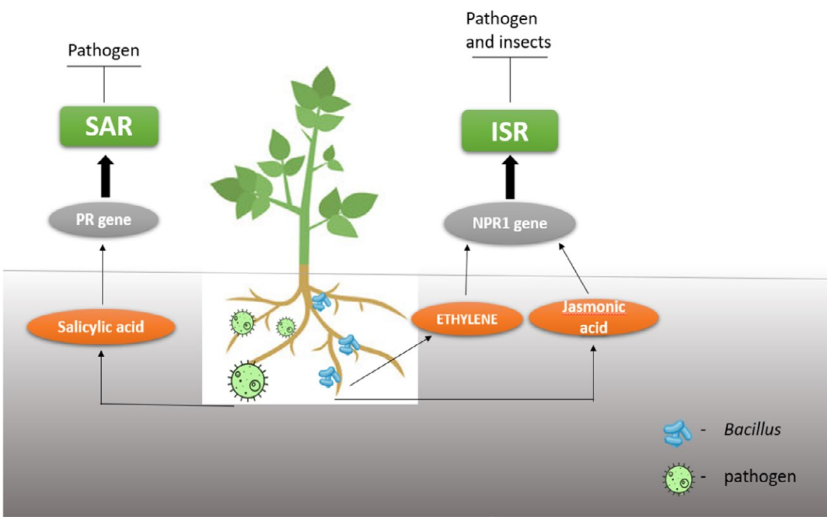

FIGURE 2 Model of induced systemic resistance (ISR) and Systemic Acquired Resistance (SAR) mechanisms elicited by *Bacillus subtilis* and pathogenic bacteria, respectively. SAR is phenotypically similar to ISR but the underlying signalling pathways are different. SAR relies on the salicylic acid signalling pathway and accumulation of pathogenesis-related proteins unlike ISR, which is elicited by the ethylene/jasmonic acid signalling pathway and regulatory gene NPR1. ISR is more diverse as compared to SAR

Rhizosphere engineering and impact on native plant endosphere and rhizosphere microbiomes

Microbes are ubiquitously present in all soil ecosystems and can be used as a strong weapon for establishing a balanced agricultural system (Köhl et al., 2019). Endophytic bacteria are the natural inhabitants of the plant, owing to their growth and development benefits (Lastochkina et al., 2020). Root endophytic communities are different from the rhizospheric community and they have a crucial role in regulating plant metabolism. The invasion of exogenous species modifies it in a context-specific manner. A recent experiment using UniFrac measurement of beta diversity proved that inoculation of *Bacillus* species into the endophytic bacterial community has an impact on the evenness, diversity and composition (Gadhave et al., 2018). Due to interspecific competition, a reduction of *Pseudomonas* species occurred due to *Bacillus* spp. All these experiments were conducted using high-throughput metagenome sequencing and assigning OTUs to each bacterial taxa. Inoculation of biofertilizer containing *Bacillus* sp. in sweet potato can increase the Shannon diversity index (H) by increasing the number of genera (alpha-proteobacteria, flavobacteria) present before inoculation (Figure 3; Salehin et al., 2020). In the soybean root rhizobia where there is an abundance of alpha and beta proteobacteria, invasion of other microbes has shown to increase microbial diversity (Zhang et al., 2011). This change in endophytic bacterial diversity has a correlation with root nodule formation and is predicted to influence the soybean plant growth in a positive way.

Rhizoremediation is an approach to deal with polluted environments in an eco-friendly manner (Alotaibi et al., 2021). Sometimes inoculation of endophytic bacteria with specific plants affects the endophytic bacterial community of the plant either positively or negatively. Inoculation of endophytic *Bacillus cereus* ERBP into *Euphorbia milii* for formaldehyde removal can modify the natural endophytic diversity (Khaksar et al., 2016). *Bacillus subtilis* BSn5, an endophytic bacterium, isolated from *Amorphophallus konjac*, can evade the plant defense response for its colonization. It produces a lantibiotic, subtilomycin to bind with self-produced flagellin resulting in endophytic colonization in *Arabidopsis thaliana* (Deng et al., 2019).

The application of *Bacillus* spp. as a plant inoculant affects various attributes of the microbial community (Gadhave et al., 2018). As the endophytic bacterial community is a principal determinant of plant growth and development, their change in nature and diversity could affect the beneficial bacteria and plant health. In general, the introduction of *Bacillus* spp. leads to an increase in the beta indices signifying the presence of some new genera. Those genera might be present in the soil at a very low density before, but they could not flourish due to some environmental or genetic factors. The introduction of *Bacillus* may pave the path for their enrichment. Thus, the development of a modified bacterial community may affect plants positively or negatively depending upon several parameters. The interspecific competition varies according to the geographical area, crop variety, and soil texture.

Soil is the hotspot for all microbial communities. Its management has a great impact on the overall microbiota. *Bacillus subtilis* produces both ribosomal and non-ribosomal peptides (NRP) as secondary metabolites, which has a great impact on indigenous microbes (Kiesewalter et al., 2020). The interactions of micro-organisms were evaluated by setting up a soil-derived semi-synthetic mock community supplemented with NRP-deficient strain *B. subtilis* WT P5_B1. 16S amplicon sequencings revealed that NRP-producing strains reduced the abundance of two genera of bacteria or had no impact on other microbial species. Engineering of *Bacillus* strains with no effect on endogenous microbiota would be beneficial for both plant and soil ecosystems.

Plant roots associate with an enormous number of microbes that shape the plant's structural and functional characteristics. The complex root microbial community is referred to as the second genome of the plant. Thus, plant rhizosphere competence is a critical prerequisite for the successful use of PGPR (Latif et al., 2020). Rhizosphere–PGPR interactions vary with plants, seasons and the texture of the soil. Actinobacteria, Acidobacteria, Proteobacteria, Chloroflexi, Bacteroidetes and Firmicutes are the abundant bacterial groups found in the rhizospheric soil. Inoculation of exogenous bacteria has shown their up- and downregulation, influencing plant health. Therefore, inoculation of PGPR should be done with a complete analysis of soil characteristics. Extensive research has been carried out to scrutinize the impact of exogenous *B. subtilis* upon the rhizospheric microbial community. The invasion of *Bacillus* has a transient effect on the bacterial community, whereas a longer effect on Eukaryota is shown in the tomato plant (Qiao et al., 2017). In the case of chamomile, there is a shift in microbial community structure and beta indices, thereby adding new functionalities to plant metabolome (Schmidt et al., 2014). However, there was no quantitative difference in microbial communities. In some cases, the impact of the introduction of *Bacillus* is soil specific that is the population density of *Bacillus* in clay soil had a significant negative correlation on bacterial diversity in cucumber rhizosphere in comparison to loam and sandy soils (Li et al., 2016). *Bacillus* inoculation also increased the aerobic stability and fermentation quality of crops. However, it does not transfer genes horizontally to soil bacteria, which is a positive side attribute.

The introduction of *B. subtilis* to tobacco rhizospheric soil showed an opposite trend (You et al., 2014). Based on the construction of a 16S rRNA gene clone library and amplified ribosomal DNA restriction analysis (ARDRA), it was inferred that *B. subtilis* inoculation increased the numbers of γ-Proteobacteria and α-Proteobacteria but reduced the numbers of bacterial groups such as β-Proteobacteria, Planctomycetes and Firmicutes (Table 3). Similarly, inoculation of *B. subtilis* Tpb55 in the tobacco rhizosphere decreased Actinobacteria level and increased the abundance of Acidobacteria and Proteobacteria (Han et al., 2016). Apart from the interaction among different bacterial groups, the inoculant can also affect the species richness of different microbial communities inhabiting the soil rhizosphere. The addition of *B. amyloliquefaciens* NJN-6 has been shown to decrease the alpha diversity of the fungal community (Yuan et al., 2017). *Bacillus subtilis* 50-1 addition to *Panax ginseng* rhizosphere decreased the bacterial diversity and increased the fungal diversity (Dong et al., 2018). Thus, the microbial community interaction varies with the plant and microbial species composition of the rhizosphere.

Plant-microbe cross-signalling

Interaction between plant and *B. subtilis* is a complicated interwoven system. The integration and fine-tuning of signals produced by PGPR affect the local or distant plant parts at the molecular and biochemical levels. To address these interconnecting links, multi-omics approaches can be used at genetic, proteomic, transcriptomic and metabolomic levels in plants (Figure 4; Meena et al., 2017). Furthermore, these techniques can be used for finding new resources with new genetic makeup for transgenic and hybrid plant technology (Parray & Shameem, 2020). Metagenomics, meta-transcriptomics, meta-metabolomics and meta-proteomics are the evolving technologies to understand the biological underpinnings of microbial communities inoculated with *B. subtilis*. Integration of these technologies with bioinformatics and statistics will help to reveal the microbial correlation and functional machinery (Ramakrishna et al., 2019). Rhizosphere engineering based on the knowledge of signalling pathways would optimize plant health and production.

Abiotic stress responses can be characterized by metabolomics as the metabolomes of the plant and rhizospheric microbiota change according to external stress responses. These can be used to dissect the tripartite (plant-beneficial microbe [*B. subtilis*])-toxicogenic microbe) interaction in agroindustrial research (Adeniji et al., 2020). Toxins generated by plant phytopathogens like *Alternaria* spp., *Penicillium* spp., *Fusarium* spp. have a negative impact on plants. Metabolomic studies revealed catabolite repression of poly-gamma-glutamic acid (PGA) produced by *Bacillus licheniformis* ATCC 9945, which has several applications in food and cosmetics industry (Mitsunaga et al., 2016). Furthermore, massive genetic mapping and data analysis can provide a biochemical solution to molecular pathology. Genome sequencing and comparative analysis of two *B. subtilis* strains, TR21 and R31 has shown several phytobeneficial characteristics like sporulation, antibiotic synthesis, and quorum sensing for better rhizosphere engineering (Li et al., 2021). Epigenomics can boost our knowledge and understanding of the below-ground dynamics through the study of methylation of DNA, alteration of histones and small non-coding RNAs. DNA methylation induced the expression of callose deposition and *AtPR1*

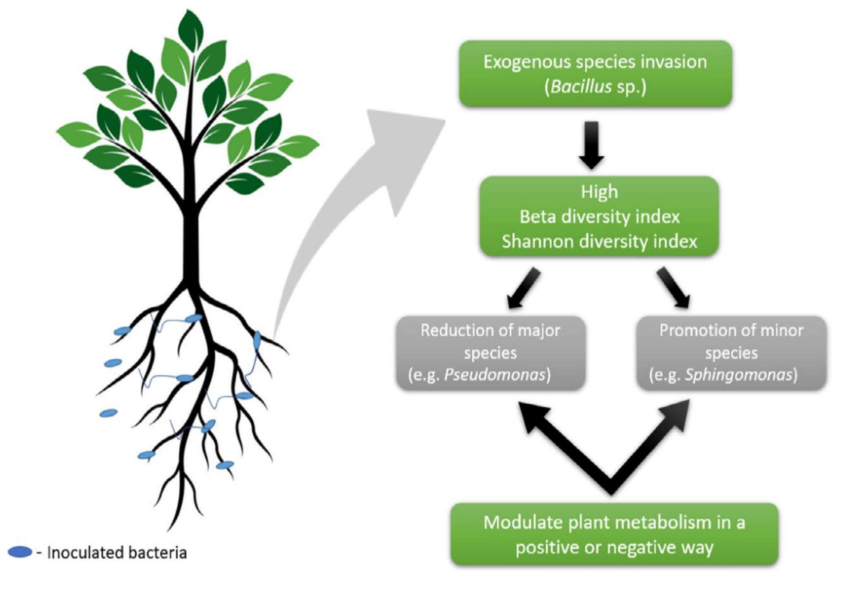

FIGURE 3 Impact of exogenous *Bacillus* spp. invasion on endophytic microbiota. Inoculation of *Bacillus* spp. increases the beta diversity index and Shannon diversity index showing a disbalance in species diversity. These variations simultaneously affect the plant physiological mechanisms positively or negatively depending upon several parameters

gene in plants challenged with oomycetes pathogens (Zogli & Libault, 2017). Similarly, epigenetic modification regulates the expression of NONEXPRESSOR OF PR GENES1 (NPR1/NIM1), a key player in SAR, thereby conferring enhanced protection against pathogens (Singh et al., 2014). Furthermore, chromatin assembly factor-1 (CAF-1) modulates histone trimethylation of H3K4 and the expression of genes encoding PR5, PR1, WRKY53 and WRKY6 involved in defence responses against biotic stresses. The genome sequencing of *B. amyloliquefaciens* FZB42 identified genes for the synthesis of secondary metabolites associated with its survival in soil and competing with neighbouring rhizospheric microbial community through the action of bacillibactin, lipopeptide and polyketides (Chen et al., 2007). The inoculation with *Bacillus* spp. altered the composition and abundance of miRNA in *Arabidopsis thaliana* root (Xie et al., 2017). The differentially expressed miRNA (miR159b, miR400 and miR8167a) enhance the expression of plant defence-related genes and polyamine biosynthesis genes associated with root growth, fruit development and cell division. The small RNAs can be used in 'cross-kingdom RNAi' for gene silencing in trans-position in host–microbe interaction to control pathogenesis and crop production improvement (Huang et al., 2019).

IMPROVEMENT OF SOIL QUALITY

Soil quality and fertility are defined by physical, chemical and biological properties of soil (Zhao et al., 2018). These include soil organic carbon, pH, macro- and micronutrients required by plants, microbiota, soil enzymes, etc.

Biofertilizers

Studies have been carried out for a better understanding of the plant growth potential of *B. subtilis* based on genome sequencing, where strain EA-CB0575 increased the total dry weight (TDW) of tomato plants as compared to non-inoculated plants in a greenhouse study, which was associated with the presence of strain-specific genes and metabolic pathways (Franco-Sierra et al., 2020; Posada et al., 2018). It is crucial to understand the plant–microbe interactions and root colonizing ability of the bacteria before using it as a biofertilizer. A healthy plant–*Bacillus* relationship would be beneficial for the plant for sustainable agriculture (Blake et al., 2021). Furthermore, zinc nanoparticles constructed using *B. subtilis* enhanced shoot length, root length and protein content in *Zea mays* (Sabir et al., 2020). Similarly, siderophore production and biofilm formation are essential requirements for iron homeostasis in *Bacillus subtilis* (Rizzi et al., 2019). The use of biofilm-forming PGPR with novel scientific approaches in agriculture would provide a big leap for the biofertilizer industry. Several studies have proved the effectiveness of PGPRs for plant growth promotion, phytohormone synthesis, nutrient uptake, systemic resistance and abiotic stress (Ramakrishna et al., 2019; Ramakrishna et al., 2020). PGPB addition to marginal land helps in the improvement of land productivity while influencing carbon cycling (Ramakrishna et al., 2020). However, if PGPRs act for their own existence and benefit, they are likely to influence the existing microbiome of the plant. In close proximity, bacteria need to cooperate for nutrients and space while developing a dynamic communication network like quorum sensing, quorum quenching and production of several secondary metabolites (Mielich-Süss & Lopez, 2015). The rhizospheric C:N ratio shapes the microbial diversity under different concentrations of atmospheric CO_2. The availability of nitrogen and nutrients favours the sequestration process along with plant growth (Grover et al., 2015). Different endospore-forming *Bacillus* strains were characterized for suppressing plant pathogens and stimulating plant growth and development. *Bacillus amyloliquefaciens* FZB42 is successfully commercialized as a biofertilizer by ABiTEP GmbH. Similarly, it was reported that biofertilizers comprising of charcoal mixed *B. subtilis* and *Azotobacter chroococcum* entrapped in organic matrix, cow dung rice bran, dried powder of neem leaves and clay named as super granules of biofertilizers (SGBF) significantly improved wheat growth and yield (Kumar et al., 2015). *Azospirillum*, *Bacillus megatherium* var. Phosphoticum, *Frateuria quaratia*, *Thiobacillus thiooxidans* and mycorrhiza combined with 100% dose of recommended fertilizers treatment significantly improved grain yield, nutrient content and biomass of the rice landrace *Padmarekha* (Nataraja et al., 2021).

Though there are several beneficial outcomes of using *B. subtilis* and *Bacillus* spp. in the agroecosystem, certain limitations exist for their use as a biofertilizer. Due to the production quality, shelf life, competence in field conditions and lesser efficiency as compared to chemical fertilizers, biofertilizers are not able to reach farmer's satisfaction (Pathania et al., 2020). Lack of trained persons and production facilities fail to produce quality and quantity of required biofertilizers. Storage and transport play a key role in deciding the shelf life as their improper handling decreases the microbial count resulting in inconsistent field performance. Furthermore, environmental factors like temperature, rainfall, pH, soil erosion affect the field conditions for optimal biofertilizer application (Schmidt & Gaudin, 2018). Chemical fertilizers show a fast response to several stress conditions while PGPR association with plant roots is time-consuming which makes the farmers

TABLE 3 Impact of exogenous *Bacillus* sp. inoculation on rhizosphere microbial community

Bacillus sp.	Plant	Effect	Reference
subtilis	*Brassica campestris*	Increased Proteobacteria and Actinobacteria	Liu et al. (2018)
amyloliquefaciens NJN-6	In vitro study	Decrease alpha diversity of fungal community	Yuan et al. (2017)
subtilis 50-1	*Panax ginseng*	Bacterial diversity decreased and fungal diversity increased	Dong et al. (2018)
subtilis	*Medicago sativa*	Bacterial diversity increased and transformation of microbial diversity composition	Li et al. (2021)
subtilis PTS-394	Tomato	Longer impact on eukaryota than bacterial community	Qiao et al. (2017)
subtilis Co1-6	Chamomile	Shifting of microbial community structure and beta indices; added extra functionalities to plant metabolism	Schmidt et al. (2014)
subtilis B068150	Cucumber	Clay soil—negative correlation with bacteria; No effect on loam and sandy soil	Li et al. (2016)
subtilis	Tobacco	γ-Proteobacteria and α-Proteobacteria increased; β-Proteobacteria, Planctomycetes, and Firmicutes decreased	You et al. (2014)
subtilis	Cotton	Quantity and species richness of soil bacteria increased (mostly actinomycetes)	Lyu et al. (2019)
subtilis Tpb55	Tobacco	Species enrichment; Actinobacteria levels decreased; Proteobacteria and Acidobacteria increased	Han et al. (2016)
--	Maize	High bacterial diversity (Actinobacteria, Bacteroidetes, Acidobacteria, Chloroflexi)	Chaudhary et al. (2021)

bypass the use of biofertilizers. But, once the rhizospheric interlink forms, it is long-lasting and can be extended to other crop plants.

Nitrogen cycling

The growing food demand in the current world has mediated the farmers to use large amounts of chemical fertilizer to increase the crop yield. The use of excessive nitrogen fertilizer imposes a serious threat to the global soil nitrogen cycle. Between 1961 and 2007, nitrogen fertilizer intake and crop yield increased, but the efficiency of nitrogen remained at 40% (Stevens, 2019). Increasing ammonia concentration in air leads to the formation of regional haze, blurred vision as well as problems associated with respiratory and cardiovascular systems. The use of *B. subtilis* as a biofertilizer reduced the percentage of ammonia emission by 44% (Sun et al., 2020a). This bacterium downregulated the *UreC* gene expression and upregulated several functional genes involved in ammonia oxidization. During composting manure, inoculation of *B. subtilis* with two other micro-organisms significantly reduced the ammonia nitrogen while increasing the nitrification process. This promotes composting technology by reducing odour emissions.

Nitrogen loss from agricultural land occurs through two pathways; leaching loss and nitrogen runoff (Mahmud et al., 2021). The released nitrogen causes groundwater contamination and eutrophication of surface water. The deposited form contributes to the global biodiversity loss in the ecosystem and soil acidification. Substitution of 50% urea with *B. subtilis* biofertilizer can reduce the loss of nitrogen by 54%. The biofertilizer downregulates the nitrogen-fixing nif H and amoA genes while upregulating the denitrifying *narG, nirS, nirK* and *nosZ* genes in the soil (Sun et al., 2020b). The successful application of *Bacillus* species in the purification of nitrogen-contaminated wastewater has been shown in several parts of the world (Hlordzi et al., 2020). There are several methods of nitrate removal like reverse osmosis, electrochemical reduction, ion exchange and electrodialysis, which are expensive. The use of conventional methods like bioaugmentation of wastewater with microbial agents in combination with the bioelectrochemical process speeds up denitrification (Rahimi et al., 2020). This can be used for the commercial production of butyrate from carbohydrate-containing wastewater like dairy farm discharges. The use of *B. subtilis* JD014 has been shown to be efficient in the removal of NO_3-N and NO_2-N along with a higher tolerance for nitrogen toxicity (Yang et al., 2021). This also helps in nitrogen fixation by purple non-sulphur bacteria, *Rhodopseudomonas palustris* by nifH gene expression and increasing oxygen uptake for nitrogenase activity (Arashida et al., 2019). Thus, *B. subtilis* biofertilizer could

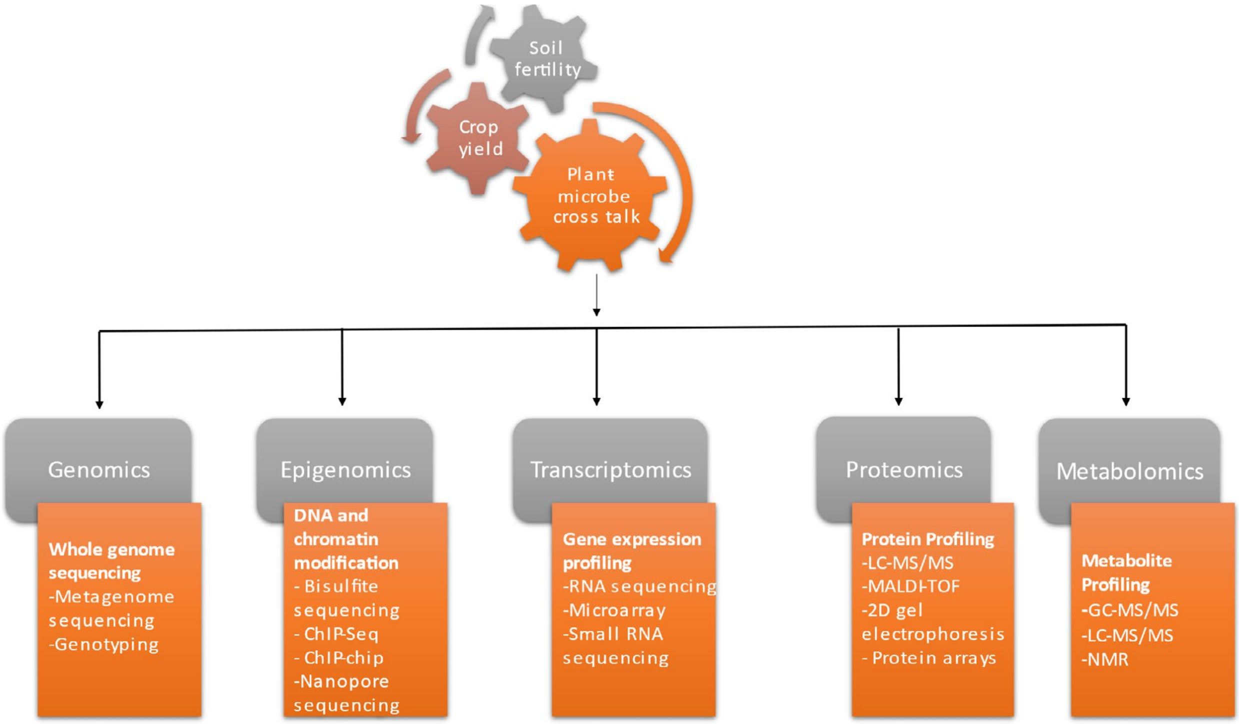

FIGURE 4 Omics approaches to identify correlation among plant-microbe cross-talk, crop yield and soil fertility. It is essential to understand the basic genetic makeup, DNA modifications, gene and protein expression patterns and metabolite dynamics for optimization of plant growth and maintenance of soil microbiome

serve as a potent fertilizer for the microbial transformation of nitrogen and reduce its loss from agroecosystems.

In the modern era of technology, *B. subtilis* MTCC1455 has been shown to be useful in the production of green cosmetics. Nitrate induces the production of surfactin which in combination with bioactive essential oils, form stable nanoemulsion, which can be stored for up to three months (Ganesan & Rangarajan, 2021). Instead of using toxic products which have several of side effects, the use of such green cosmetics would be effective and beneficial for mankind.

Bioremediation of pollutants

Industrialization and technological advances have led to a world full of hazardous substances, toxic metals, organic and inorganic pollutants leaving a harmful impact on the ecosystem (Ferronato & Torretta, 2019). To purify the soil from these toxic metals, steps have been taken by various governments (Wuana & Okieimen, 2011). However, the use of several artificial substances and physiochemical processes has greater side effects on living organisms. Bioremediation is an effective eco-friendly technology to purify heavy metal contaminated soil, where bacteria play a vital role. Soil is the reservoir of microbes and they regulate the soil ecosystem by maintaining several biogeochemical cycles of carbon, nitrogen and different toxic metals. They have the potential to mobilize and immobilize the metals as well as decide their fate in different geological forms. *Bacillus subtilis* has been shown as a potent micro-organism in remediating soil contaminated artificially with chromium (Li et al., 2019). It can be used as a potent geomicrobiological stabilizer for immobilization of Pb, Sb, Ni, Cu and Zn in alkaline soils. The application of nanohydroxyapatite (NHAP) along with *B. subtilis* to cadmium contaminated soil improved soil remediation and increased rhizospheric community (Liu et al., 2018). UV-mutated species *B. subtilis* 38, served as an efficient sorbent of Hg, Cr, Cd and Pb (Wang et al., 2019). Certain species of *B. subtilis* (DBM) have been found to survive in heavy metal contaminated soils, where DBM mineral complexes were formed by phosphodiester bond formation thereby restricting the entry of heavy metals into the bacterial cells as well as increasing its Cu and Pb sorption ability (Bai et al., 2019). The pH of the soil and the density of the bacteria in the soil community have a greater impact on all these processes. *Bacillus subtilis* produce cyclic lipopeptide biosurfactants that can stabilize the pH, salinity and temperature of the soil. These biosurfactants are easily disposable, low toxic and can be prepared from low-cost agricultural by-products. They can be efficiently used for remediating oil-contaminated soils due to their excellent property of hydrocarbon emulsification (Nogueira Felix et al., 2019). Shreds of evidence suggest their effective role in the sorption of several heavy metals in different types of soil textures (Selvam et al., 2021). Bioelectrokinetic techniques have been used for the purification of crude oil hydrocarbon, which is a combination of bioremediation and electrokinetic remediation. *Bacillus subtilis* AS2 used with this technique produces biosurfactants that increase the remediation efficiency by enhancing the total organic content, which is achieved by the solubilization of hydrocarbons and faster electromigration in the anodic compartment (Prakash et al., 2021). This method can be effectively used for the purification of petroleum contaminated soils and biosorption of heavy metals from an aqueous solution. Using this bacterium with planned biosorption techniques, underground water can be purified. Microbial biochar formulations (MBFs) using biochar from microbial biowaste and *B. subtilis* SL-13 were used to treat potted pepper plants, which resulted in enhanced plant growth as well as soil fertility improvement (Tao et al., 2019).

Even after the development of effective biofertilizers, several farmlands use artificial or chemical fertilizers. As a result, the soil becomes contaminated with herbicides. Some bacteria play an effective role in the biotransformation of nicosulfuron herbicides in contaminated soils, without majorly affecting the rhizospheric microbial community (Zhang et al., 2020). Soil microbes play an important role in determining the diversity and composition of plant communities while plants impact microbial species richness, which regulates the plant metal uptake and its utilization.

Control of soil reservoirs of antibiotic resistance genes

Bacillus subtilis, the Gram-positive bacteria, has shown resistance against different antibiotics by several inbuilt physiochemical mechanisms. Recently, *B. subtilis* subsp. *subtilis* strain RK, with a novel class of aminoglycoside transferase (AHP5), responsible for resistance against aminoglycoside-based antibiotics has been identified (Parulekar et al., 2019). The cell wall transporters present in bacteria are crucial for their resistance. The BceAB type transporter present in *B. subtilis* is responsible for resistance against cell wall active antimicrobial peptide, targeting the lipid II cycle of cell wall synthesis (Kobras et al., 2020). This mechanism is often similar to the low GC content of Gram-positive microbes. Certain mechanisms where the bacteria show resistance against highly reactive oxidizing agents like chlorine dioxide and hydrogen peroxide are still elusive (Martin et al., 2015). Although it harbours several resistance genes and mechanisms, the positive traits of these bacteria have been used after removing ARGs and mobile genetic elements (MGEs).

Animal manure is used as compost for plant growth and development. Animal manure serves as a large repository of ARGs due to the heavy use of antibiotics against various animal pathogens (Zalewska et al., 2021). These ARGs can enter the food chain via plant genomes. The presence of ARGs on mobile genetic elements (MGEs) allows the spreading of genes among the microbial community through horizontal gene transfer and transposons. When they enter pathogenic bacteria, it is beneficial for the bacteria but a threat to the host organism. The addition of *B. subtilis* (0.5%) in the livestock manure used for composting can reduce the ARGs, MGEs and pathogenic bacteria by modulating pH (Duan et al., 2019). It changed the absolute gene copy numbers of the ARGs while regulating their spreading ability. The use of mature compost during swine manure composting decreased the relative abundance of ARGs, especially sulphanilamide ARGs and MGEs while promoting organic matter decomposition. This is due to the rapid growth of *Bacillus* and *Thermobacillus* in the thermophilic period of composting and the reduction of bacterial hosts (Firmicutes) for ARG and MGEs (ISCR1 and int1) (Wang et al., 2020).

On the contrary, *Bacillus* sp. was reported to be the major carrier of ARGs based on a study where a microbial agent was used with pig manure for compost, which resulted in decreased levels of four categories of ARGs along with Firmicutes and Tn916/1545 while two categories increased their levels by amplification of intI1 and several host bacteria (Cao et al., 2020). Although it is difficult to conclude whether *B. subtilis* acts as a bacterial host for spreading ARGs or decreasing their abundance, by affecting other hosts, as it changes its nature in different circumstances, we can infer that its positive or negative impact depends on the nature, chemical composition of the manure, composting process and the concentration of bacteria used in the process.

Carbon sequestration

CO_2 constitutes 76% of the total Greenhouse gas emission and reducing its level in the atmosphere is one of the most challenging issues faced by mankind (Rossi et al., 2016). Biological systems play a crucial role in alleviating higher CO_2 concentration through carbon sequestration. Plants allocate about one-third of their photosynthetically fixed carbon to soil (Jones et al., 2009). It is the largest reservoir of several minerals. The soil carbon sequestration is 3.5-5.2 Gt/year, which needs to be increased to mitigate the recent issue of high CO_2 levels leading to global warming (Duran et al., 2021). The soil microbiota nourished by root exudates regulates the carbon efflux and influx pumps in the soil rhizosphere. The treatment of drylands with micro-organisms from biological crust increases the carbon adsorption capability of soil from 0.232 to 0.294 g/m^2/day (Kheirfam, 2020).

Bacillus subtilis, a common PGPB used in the agricultural sector has the potential for employment for carbon sequestration in various growth conditions. Several experiments have been carried out to check the carbon sequestration ability of *Bacillus*. Carbonic anhydrase (CA) based enzymatic technology is an approach for carbon fixation and its biomitigation (Effendi & Ng, 2019). The immobilized forms are efficient and can be used for industrial purposes. The CA isolated from *B. subtilis* VSG4 convert CO_2 into $CaCO_3$ in the immobilized form (Oviya et al., 2012). *Bacillus* sp. SS105, isolated from free-air CO_2 enriched (FACE) condition, can transform CO_2 to calcite form due to the presence of beta- and gamma-carbonic anhydrase genes. The CO_2 sequestration efficiency and calcite formation were observed by RUBISCO and CA enzyme assays (Maheshwari et al., 2019). This sequestration efficiency of bacteria also helps in the maintenance of several physiological and biological functions. The lipopeptide type biosurfactants are also produced. Simultaneous carbon sequestration and biosurfactant production by *Bacillus* would be beneficial from both commercial and agricultural points of view (Maheshwari et al., 2017). Gaseous CO_2 and $NaHCO_3$ are good alternatives to produce biosurfactants using *Bacillus* sp. ISTS2 (Sundaram & Thakur, 2015).

Although the use of *Bacillus* spp. for carbon sequestration is an efficient method, it has some side effects with reference to manure composting. The production of organic matter, humus or carbon conversion has a crucial effect on plant growth and development. *Bacillus subtilis* applied at lower concentration (0.5%) reduced mineralization in the cooling and maturity stages of composting and enhanced the humification of carbon (Duan et al., 2020). However, inoculation of *B. subtilis* at a higher percentage (>2%) increased the fermentation process but inhibited the synthesis of total organic carbon and humus (Lu et al., 2021). Thus, the concentration of the bacteria used for composting affects the sequestration ability of the soil. It is important to consider these factors while applying *Bacillus* as a biofertilizer.

Although *B. subtilis* is the common PGPR in use, there are some dark sides akin to Mr. Hyde character that need to be explored. It is the major carrier of antibiotic resistance genes and shows resistance against various pesticides. Though horizontal gene transfer has not been discovered yet, it may affect the underground microbial diversity as it can activate several other transposons for gene transfer. More research is required to gain a deeper insight into it. The concentration of *B. subtilis* has to be considered while applying as a biofertilizer. It can modulate the biogeochemical cycles, carbon sequestration process, and humus production when applied at a higher concentration. The soil texture, composition, pH, temperature, climatic condition are some crucial factors that affect the bacteria association with plants and their efficiency. Before bacteria inoculation, the history of the farmland should be checked as the effect of *Bacillus* on the rhizospheric microbial community changes with the soil variety and texture. Some soil types do not affect the microbiome while a different soil type decreases the microbial diversity.

CONCLUSION

Though there is a high demand for food production in the current scenario, a focus on sustainable agricultural practices without harming the environment. PGPR, frontbenchers of the next-generation green revolution, are good alternatives to chemical fertilizers. *Bacillus subtilis* is one of the common PGPR to be used efficiently for optimization of plant growth and yield. It protects the plant from several stress conditions by ISR, biofilm formation, lipopeptide, siderophore and exopolysaccharide secretion. It acts as an efficient denitrifying agent in the agroecosystem and maintains soil health by eco-friendly remediating technologies.

Future perspectives

Future research should be focused on several antibiotic resistance genes present in *Bacillus* whose transfer and interaction with other microbes and the effect on soil microbial diversity. Development of microbial consortia with *B. subtilis* for enhancement of crop yield without any negative impact on the rhizosphere microbial community is the need of the hour. Targeting the endophytic microbiome of the seed, selecting plants compatible with the microbial inoculant or employing microbial engineering methods which ensure the continuity of beneficial microbiota for generations are some options for the efficient deployment of *Bacillus subtilis* and other PGPB for green agriculture. Soil amendments and plant genome modification would help maintain soil quality. Metagenomics, meta-transcriptomics and epigenomics are several emerging technologies that can be used for functional assessment of crop, soil and climatic factors for better integration with the traditional system of agriculture. All these steps can enhance productivity and sustainability in diverse agro-ecological conditions. Overall, the same *B. subtilis* akin to the character of Dr. Jekyll and Mr. Hyde can have a positive or negative effect based on the density of bacteria applied in the field, interactions with soil microbial community, soil characteristics, environment and plant species. The state-of-the-art research would help in eliminating the negative attributes of *Bacillus subtilis* and other PGPB on plant growth, soil health and environment.

CONFLICT OF INTEREST

No conflict of interest to report for all authors.

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